One-variable data: Distributions and measures of center and spread

Problem-Solving and Data Analysis · Math · 84 questions

1 a214fab5-1eb4-49e2-ad3e-2b74eb180945 · Easy

For the last three years, a pig at a farm gave birth to one litter per year. The table shows the number of piglets in each of these three litters.

Litter number	Number of piglets
1	8
2	7
3	9

What is the mean number of piglets per litter for these three litters?

A. 2

B. 7

C. 8

D. 9

2 862ae133-b6b4-494f-a4f0-2a3ec7706432 · Easy

Shape	Number of cards
Square	5
Circle	9
Triangle	14
Star	12

The table summarizes the distribution of shape printed on one side for cards in a deck of cards. How many cards in the deck have a triangle printed on one side?

Student-produced response: _____

3 031602-DC · Easy

Five Smallest Countries in 2016

Country	Land area (square kilometers)
Monaco	2.0
Nauru	21
San Marino	61
Tuvalu	26
Vatican City	0.44

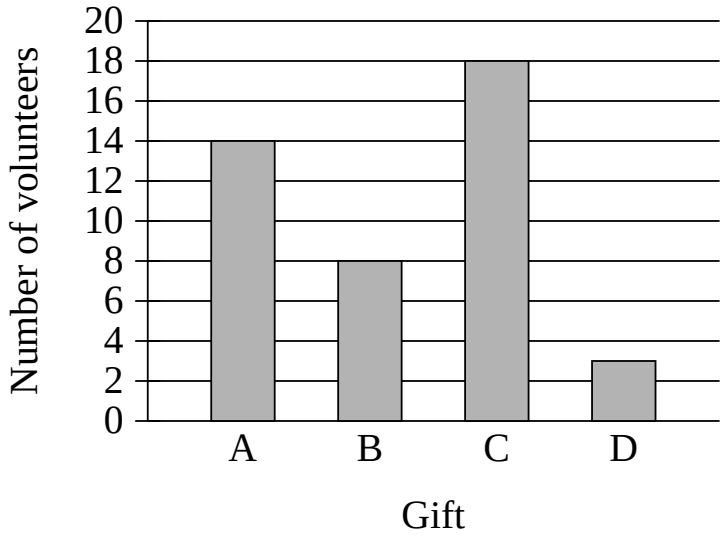
The table above shows the land area, in square kilometers, of the five smallest countries of the world in 2016. Based on the table, what is the mean land area of the 5 smallest countries in 2016, to the nearest square kilometer?

A. 20

B. 22

C. 61

D. 110



- The Number of volunteers data for the 4 bars are as follows:
  - A: 14
  - B: 8
  - C: 18
  - D: 3

In April, there were 43 volunteers in a cleanup project. Each volunteer was asked to choose a small gift labeled A, B, C, or D. The bar graph shows the number of volunteers who chose each gift. How many volunteers chose gift C?

A. 3

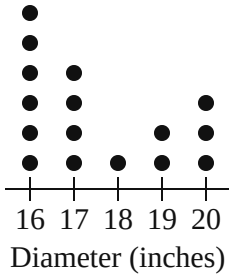
B. 8

C. 14

D. 18

5 32c749f0-91ae-4721-8ecd-9eac7aa1f372 · Easy

To study the characteristics of sea stars in a group of tide pools, researchers measured the diameter of the sea stars within the tide pools. The dot plot gives the diameter, to the nearest inch, of each of the sea stars in these tide pools.



- The data for the dot plot are as follows:
  - 16: 6 dots
  - 17: 4 dots
  - 18: 1 dot
  - 19: 2 dots
  - 20: 3 dots

Based on the dot plot, how many sea stars had a diameter, to the nearest inch, of 16 inches?

A. 16

B. 6

C. 4

D. 1

6 43a2c649-5ad6-4d16-9779-da333ff1569a · Easy

2, 9, 14, 23, 32

What is the mean of the data shown?

A. 14

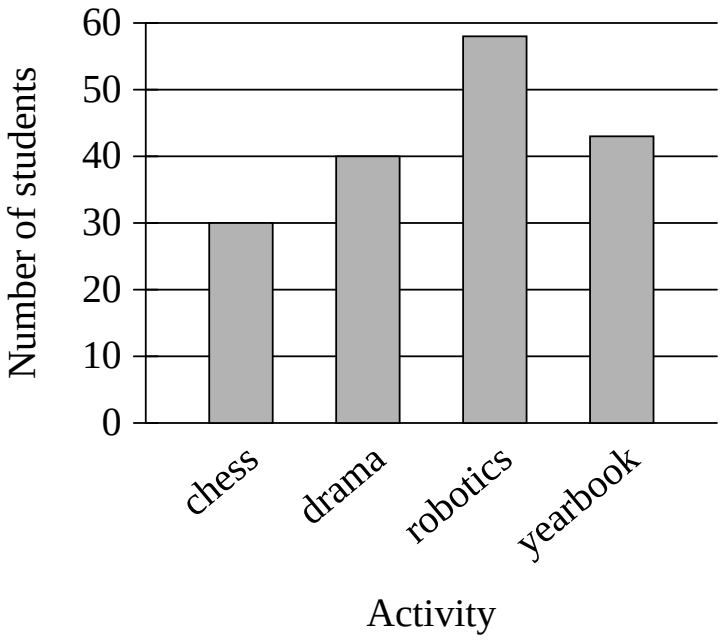
B. 16

C. 17

D. 32

7 b00de09c-a654-4119-9b48-e0de375ac094 · Easy

The bar graph shows the distribution of the number of students in each of four extracurricular activities at a high school.



- The Number of students data for the 4 bars are as follows:
  - chess: 30
  - drama: 40
  - robotics: 58
  - yearbook: 43

How many more students are in drama than in chess?

A. 10

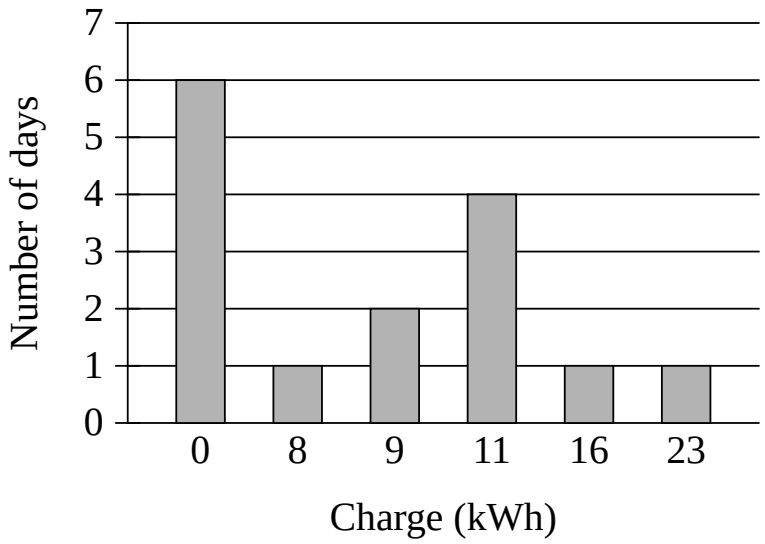
B. 30

C. 40

D. 70

8 f5598292-13c4-4d60-8b84-b381d3c86f9c · Easy

The bar graph summarizes the charge, in kilowatt-hours kWh, a battery received each day for 15 days.



- The data for the 6 categories are as follows:
  - 0 kilowatt-hours: 6
  - 8 kilowatt-hours: 1
  - 9 kilowatt-hours: 2
  - 11 kilowatt-hours: 4
  - 16 kilowatt-hours: 1
  - 23 kilowatt-hours: 1

For how many of these 15 days did the battery receive a charge of $0 \ \ \mathrm{kWh}$?

A. 0

B. 1

C. 4

D. 6

9 db1f16a5-024c-473c-aedc-cdda8967d1c1 · Easy

$$73, 74, 75, 77, 79, 82, 84, 85, 91$$

What is the median of the data shown?

Student-produced response: _____

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10 88659c2c-a185-4a2b-b844-b4bdc689c709 · Easy

Task	Time (minutes)
I	5
II	2
III	15
IV	9
V	9

The table shows the amount of time it took Robert to complete each of the 5 tasks in a project. What was the mean time, in minutes, for Robert to complete a task for this project?

Student-produced response: _____

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11 5028e152-9c24-4fde-a87c-5eb62669957e · Easy

4, 4, 4, 4, 8, 8, 8, 13, 13

Which frequency table correctly represents the data listed?

A.

Number	Frequency
4	4
8	3
13	2

B.

Number	Frequency
4	4
3	8
2	13

C.

Number	Frequency
4	16
8	24
13	26

D.

Number	Frequency
16	4
24	8

26	13

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12 abb8d90e-7d43-41be-9dae-ad5e692240f5 · Easy

Data set X: 5, 9, 9, 13

Data set Y: 5, 9, 9, 13, 27

The lists give the values in data sets X and Y. Which statement correctly compares the mean of data set X and the mean of data set Y?

A. The mean of data set X is greater than the mean of data set Y.

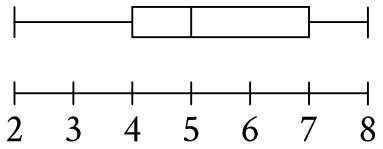
B. The mean of data set X is less than the mean of data set Y.

C. The means of data set X and data set Y are equal.

D. There is not enough information to compare the means.

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13 da3d86e5-c459-4887-9ed0-fecf17d87773 · Easy



- From left to right the values of the vertical bars in the box plot are as follows:
  - First vertical bar: 2
  - Second vertical bar: 4
  - Third vertical bar: 5
  - Fourth vertical bar: 7
  - Fifth vertical bar: 8

The box plot summarizes 15 data values. What is the median of this data set?

A. 2

B. 3

C. 5

D. 8

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14 012a45d5-4b75-4217-b792-5b7d08ece247 · Easy

$$71, 72, 73, 76, 77, 79, 83, 87, 93$$

What is the median of the data shown?

A. 71

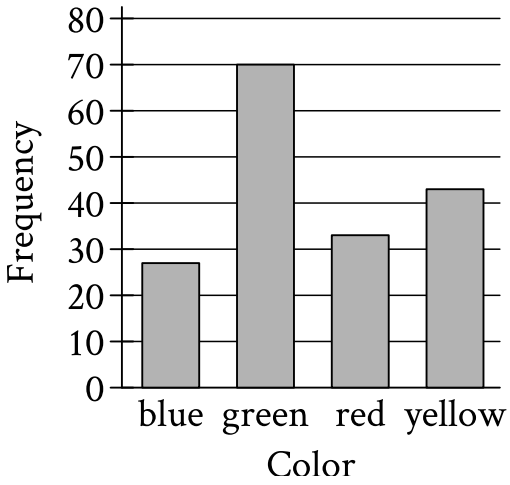
B. 77

C. 78

D. 79

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15 1010b663-23d7-48e5-8a54-02a9c7c6a552 · Easy



- The Frequency data for the 4 bars are as follows:
  - blue: 27
  - green: 70
  - red: 33
  - yellow: 43

A data set consists of 173 colors. The bar graph shows the number of times each color appears in the data set. Which color appears 70 times?

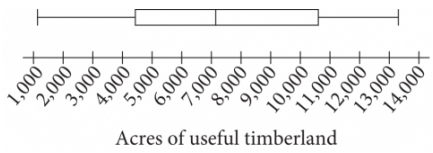
A. Blue

B. Green

C. Red

D. Yellow

16 028959-DC · Easy



The number of acres of useful timberland in 13 counties in California is summarized in the box plot above. Which of the following is closest to the median number of acres?

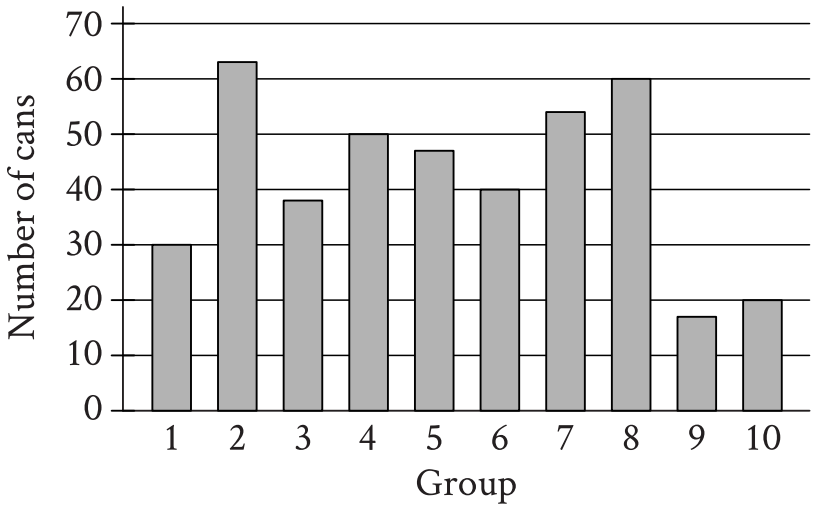
A. 4,399

B. 7,067

C. 8,831

D. 10,595

17 a6ce86a7-db0b-4ece-b1dc-2fb1ebd4b99b · Easy



- The data for the 10 categories are as follows:
  - Group 1: 30
  - Group 2: 63
  - Group 3: 38
  - Group 4: 50
  - Group 5: 47
  - Group 6: 40
  - Group 7: 54
  - Group 8: 60
  - Group 9: 17
  - Group 10: 20

The bar graph shows the distribution of 419 cans collected by 10 different groups for a food drive. How many cans were collected by group 6?

Student-produced response: _____

18 08099-DC · Easy

The high temperature, in degrees Fahrenheit (°F), in a certain city was recorded for each of 5 days. The data are shown below.

Day	1	2	3	4	5
High temperature (°F)	81	80	81	81	82

Over this 5-day period, which of the following is NOT equal to 81°F?

A. Median of the high temperatures

B. Mean of the high temperatures

C. Mode of the high temperatures

D. Range of the high temperatures

19 070909-DC · Easy

Number of High School Students Who Completed Summer Internships

High school	Year				
	2008	2009	2010	2011	2012
Foothill	87	80	75	76	70
Valley	44	54	65	76	82
Total	131	134	140	152	152

The table above shows the number of students from two different high schools who completed summer internships in each of five years. No student attended both schools. Which of the following statements are true about the number of students who completed summer internships for the 5 years shown?

1. The mean number from Foothill High School is greater than the mean number from Valley High School.
2. The median number from Foothill High School is greater than the median number from Valley High School.

A. I only

B. II only

C. I and II

D. Neither I nor II

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20 bc57ca71-eb48-4ae7-95a8-3ed00f622ed8 · Easy

Each value in the data set shown represents the height, in centimeters, of a plant.

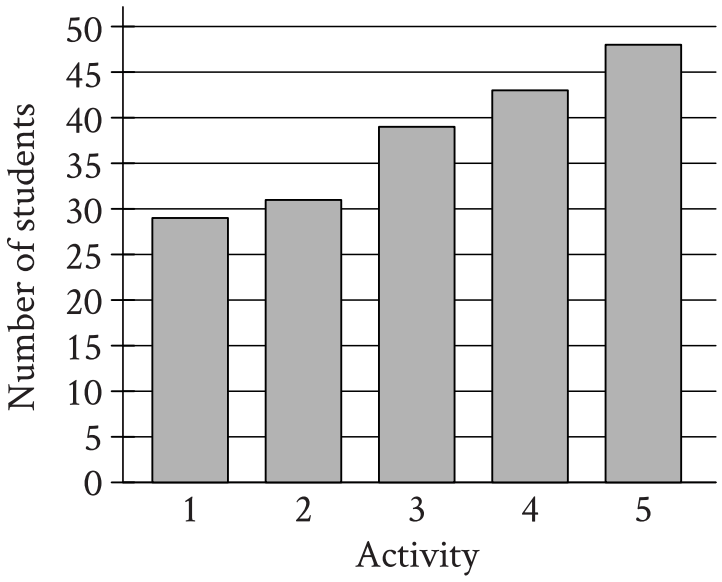
$$6, 10, 13, 2, 15, 22, 10, 4, 4, 4$$

What is the mean height, in centimeters, of these plants?

Student-produced response: _____

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21 47edb639-8ebe-471a-ae9d-172195289d2b · Easy



- The data for the 5 categories are as follows:
  - 1: More than halfway above 25 students
  - 2: Less than halfway above 30 students
  - 3: More than halfway above 35 students
  - 4: About halfway above 40 students
  - 5: About halfway above 45 students

A group of students voted on five after-school activities. The bar graph shows the number of students who voted for each of the five activities. How many students chose activity 3?

A. 25

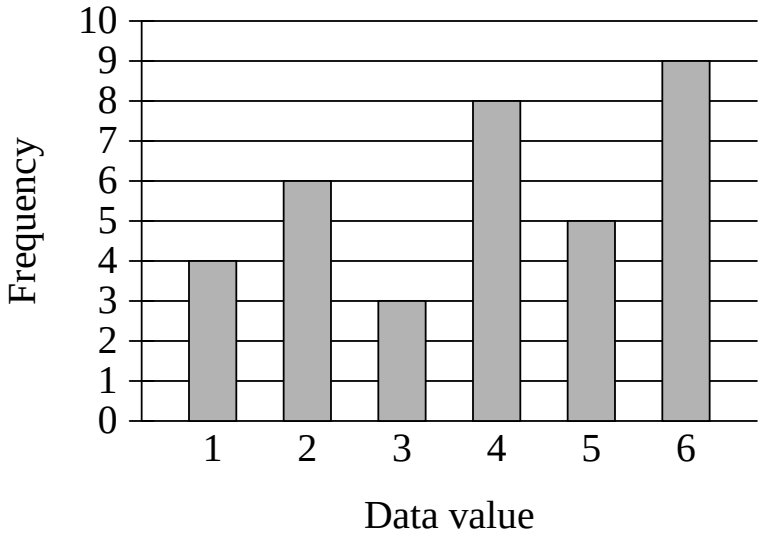
B. 39

C. 48

D. 50

22 0d175345-0c08-4bec-811b-d117627c8b48 · Easy

The bar graph shows the frequency of each data value in a data set.



- The Frequency data for the 6 bars are as follows:
  - Data value 1: 4
  - Data value 2: 6
  - Data value 3: 3
  - Data value 4: 8
  - Data value 5: 5
  - Data value 6: 9

What is the frequency of data value 2 in this data set?

Student-produced response: _____

23 027874-DC · Easy

Data set A: 1, 2, 3, 4, 5, 6, 7
Data set B: 1, 1, 2, 2, 3, 3, 4

Which of the following statements correctly compares the means of data set A and data set B?

A. The mean of each data set is 2.

B. The mean of each data set is 4.

C. The mean of data set A is less than the mean of data set B.

D. The mean of data set A is greater than the mean of data set B.

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24 026304-DC · Easy

Data set A: 72,73,73,76,76
Data set B: 61,64,74,85,x

Data set A and data set B each contain 5 numbers. If the mean of data set A is equal to the mean of data set B, what is the value of x ?

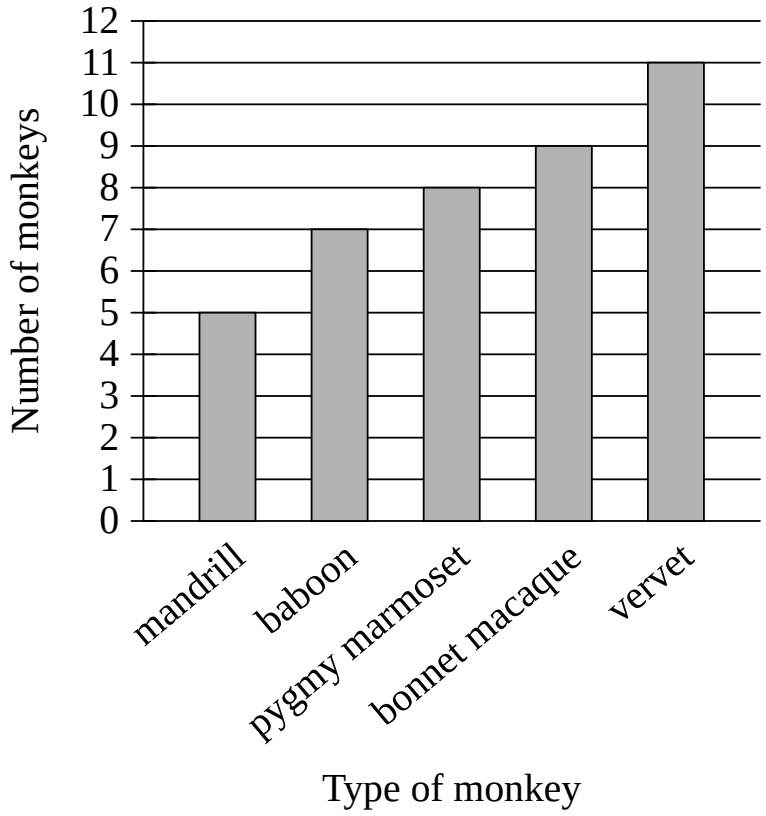
A. 77

B. 85

C. 86

D. 95

The bar graph shows the number of each type of monkey in a sanctuary.



- The Number of monkeys data for the 5 bars are as follows:
  - mandrill: 5
  - baboon: 7
  - pygmy marmoset: 8
  - bonnet macaque: 9
  - vervet: 11

How many more vervets are in this sanctuary than mandrills?

A. 11

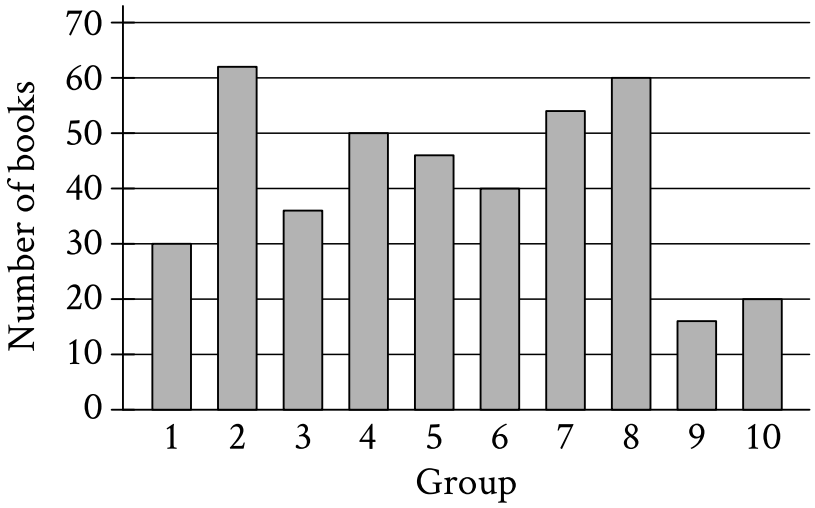
B. 6

C. 5

D. 3

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26 ede10369-b903-44c6-9ea8-c55232a55dfd · Easy



- The data for the 10 categories are as follows:
  - Group 1: 30
  - Group 2: 62
  - Group 3: 36
  - Group 4: 50
  - Group 5: 46
  - Group 6: 40
  - Group 7: 54
  - Group 8: 60
  - Group 9: 16
  - Group 10: 20

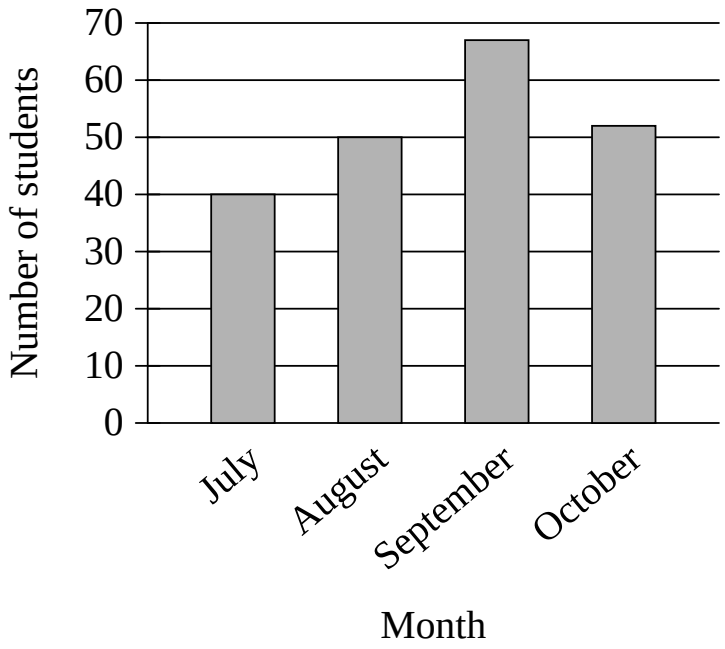
The bar graph shows the distribution of 414 books collected by 10 different groups for a book drive. How many books were collected by group 1?

Student-produced response: _____

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27 ad801a2d-5e09-4c92-b37a-9c8b027d62f2 · Easy

The bar graph shows the distribution of the number of students from one school who were born in one of four months.



- The Number of students data for the 4 bars are as follows:
  - July: 40
  - August: 50
  - September: 67
  - October: 52

How many more students were born in August than were born in July?

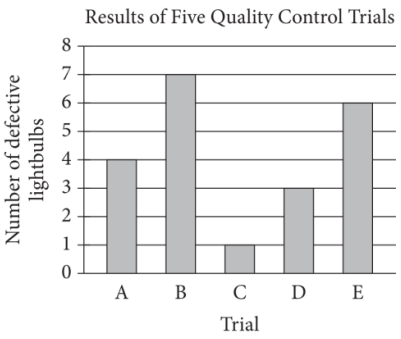
A. 90

B. 50

C. 40

D. 10

28 070910-DC · Easy



For quality control, a company that manufactures lightbulbs conducted five different trials. In each trial, 500 different lightbulbs were tested. The bar graph above shows the number of defective lightbulbs found in each trial. What is the mean number of defective lightbulbs for the five trials?

A. 4.0

B. 4.2

C. 4.6

D. 5.0

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29 ee5e0bcd-6e26-4cc7-9ca8-16eab11e3146 · Easy

6, 6, 8, 8, 8, 10, 21

Which of the following lists represents a data set that has the same median as the data set shown?

A. 4, 6, 6, 6, 8, 8

B. 6, 6, 8, 8, 10, 10

C. 6, 8, 10, 10, 10, 12

D. 8, 8, 10, 10, 21, 21

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30 a966fc8b-7920-4a60-ae82-b53184a3ba6c · Easy

Five *Eretmochelys imbricata*, a type of sea turtle, each have a nest. The table shows an original data set of the number of eggs that each turtle laid in its nest.

Nest	Number of eggs
A	149
B	144
C	148
D	136
E	139

A sixth nest with 121 eggs is added to create a new data set. Which of the following correctly compares the means of the two data sets?

A. The mean of the original data set is greater than the mean of the new data set.

B. The mean of the original data set is less than the mean of the new data set.

C. The means of both data sets are equal.

D. There is not enough information to compare the means.

31 070911-DC · Easy

A study was conducted on the production rates for a company that produces tractor wheels. The table below shows the number of wheels made during 11 consecutive one-hour production periods.

One-hour period	Number of wheels made
A	24
B	24
C	21
D	21
E	21
F	19
G	24
H	24
I	19
J	22
K	23

What is the range of the number of wheels made for the 11 one-hour periods?

A. 5.5

B. 5.0

C. 4.5

D. 4.0

32 06926-DC · Easy

For a school fund-raiser, 10 students sold a total of 90 boxes of cookies. Which of the following can be calculated from this information?

A. The average number of boxes sold per student

B. The median number of boxes sold per student

C. The greatest number of boxes sold by one student

D. The least number of boxes sold by one student

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33 8183ef2c-57c1-4464-91c3-475ac4162ca1 · Easy

Response	Frequency
Once a week or more	3
Two or three times a month	16
About once a month	26
A few times a year	73
Almost never	53
Never	29
Total	200

The table gives the results of a survey of 200 people who were asked how often they see a movie in a theater. How many people responded either “never” or “almost never”?

A. 24

B. 53

C. 82

D. 118

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34 a76daad9-3856-4e79-95ee-20158581002b · Easy

Type of store	Average number of employees
Warehouse store	365
Department store	213
Supermarket	130

For a certain region, the table shows the average number of store employees in 2016 by type of store. Based on the table, how much greater was the average number of store employees in warehouse stores than in supermarkets?

A. 83

B. 152

C. 235

D. 495

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35 026380-DC · Easy

2, 2, 2, 3, 4, 4, 11

What is the median of the seven data values shown?

A. 2

B. 3

C. 4

D. 9

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36 09906-DC · Easy

The table below shows the high and low temperatures in Houston, Texas, during a five-day period.

Temperatures in Houston, Texas
(degrees Fahrenheit)

	Monday	Tuesday	Wednesday	Thursday	Friday
High temperature	73	56	62	75	81
Low temperature	49	37	41	54	63

What was the mean low temperature, in degrees Fahrenheit, during the five-day period?

A. 48.8

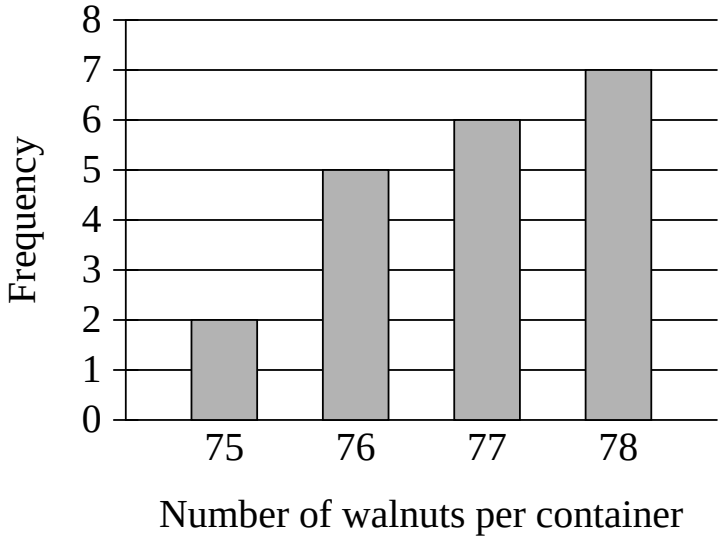
B. 49

C. 59

D. 59.1

37 11e14767-1eaf-4adb-8d85-9e42946ce439 · Easy

The bar graph shows the distribution of the number of walnuts per container for 20 containers at a grocery store.



- The Frequency data for the 4 bars are as follows:
  - 75 walnuts per container: 2
  - 76 walnuts per container: 5
  - 77 walnuts per container: 6
  - 78 walnuts per container: 7

How many of these containers of walnuts contain exactly 78 walnuts?

A. 2

B. 7

C. 20

D. 78

38 14957-DC · Medium

The results of two independent surveys are shown in the table below.

Men's Height

Group	Sample size	Mean (centimeters)	Standard deviation (centimeters)
A	2,500	186	12.5
B	2,500	186	19.1

Which statement is true based on the table?

A. The Group A data set was identical to the Group B data set.

B. Group B contained the tallest participant.

C. The heights of the men in Group B had a larger spread than the heights of the men in Group A.

D. The median height of Group B is larger than the median height of Group A.

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39 4c64fb61-5036-48e2-a518-1fe34e097aec · Medium

A list of 10 data values is shown.

$$6, 8, 16, 4, 17, 26, 8, 5, 5, 5$$

What is the mean of these data?

Student-produced response: _____

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40 11167-DC · Medium

Ages of 20 Students Enrolled in a College Class

Age	Frequency
18	6
19	5
20	4
21	2
22	1
23	1
30	1

The table above shows the distribution of ages of the 20 students enrolled in a college class. Which of the following gives the correct order of the mean, median, and mode of the ages?

A. mode < median < mean

B. mode < mean < median

C. median < mode < mean

D. mean < mode < median

41 020981-DC · Medium

The weights, in pounds, for 15 horses in a stable were reported, and the mean, median, range, and standard deviation for the data were found. The horse with the lowest reported weight was found to actually weigh 10 pounds less than its reported weight. What value remains unchanged if the four values are reported using the corrected weight?

A. Mean

B. Median

C. Range

D. Standard deviation

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42 10941-DC · Medium

Station 1	Station 2	Station 3	Station 4	Station 5
$3.699	$3.609	$3.729	$3.679	$3.729

In the table above, Melissa recorded the price of one gallon of regular gas from five different local gas stations on the same day. What is the median of the gas prices Melissa recorded?

A. $3.679

B. $3.689

C. $3.699

D. $3.729

International Tourist Arrivals, in millions

Country	2012	2013
France	83.0	84.7
United States	66.7	69.8
Spain	57.5	60.7
China	57.7	55.7
Italy	46.4	47.7
Turkey	35.7	37.8
Germany	30.4	31.5
United Kingdom	26.3	32.2
Russia	24.7	28.4

The table above shows the number of international tourist arrivals, rounded to the nearest tenth of a million, to the top nine tourist destinations in both 2012 and 2013. Based on the information given in the table, how much greater, in millions, was the median number of international tourist arrivals to the top nine tourist destinations in 2013 than the median number in 2012, to the nearest tenth of a million?

Student-produced response: _____

44 029308-DC · Medium

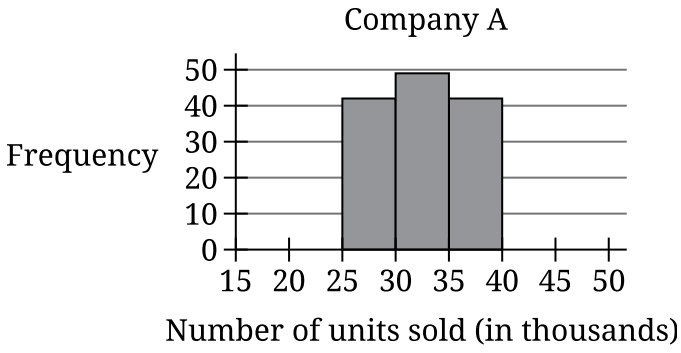
A fish hatchery has three tanks for holding fish before they are introduced into the wild. Ten fish weighing less than 5 ounces are placed in tank A. Eleven fish weighing at least 5 ounces but no more than 13 ounces are placed in tank B. Twelve fish weighing more than 13 ounces are placed in tank C. Which of the following could be the median of the weights, in ounces, of these 33 fish?

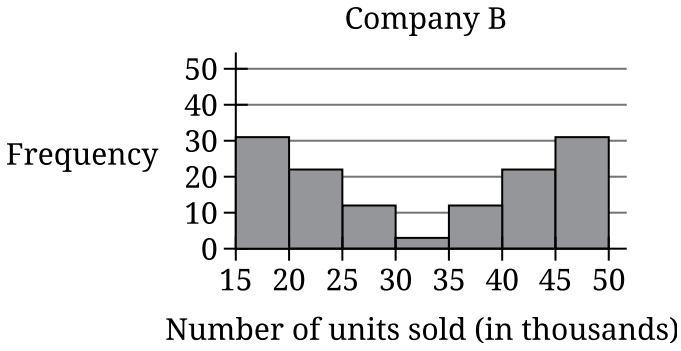
A. 4.5

B. 8

C. 13.5

D. 15





- Company A:
  - The horizontal axis is labeled Number of units sold (in thousands). It ranges from 15 to 50 and is divided into 7 equal intervals.
  - The vertical axis is labeled Frequency. It ranges from 0 to 50.
  - The histogram has 3 bins.
  - The Frequency data for the 3 bins are as follows:
    - 25 to 29: 43
    - 30 to 34: 48
    - 35 to 39: 43
- Company B:
  - The horizontal axis is labeled Number of units sold (in thousands). It ranges from 15 to 50 and is divided into 7 equal intervals.
  - The vertical axis is labeled Frequency. It ranges from 0 to 50.
  - The histogram has 7 bins.
  - The Frequency data for the 7 bins are as follows:
    - 15 to 19: 31
    - 20 to 24: 24
    - 25 to 29: 11
    - 30 to 34: 3

- 35 to 39: 11
- 40 to 44: 24
- 45 to 49: 31

The histograms summarize the distributions of number of units sold, in thousands, for company A and company B. Which statement best compares the standard deviations of number of units sold for these companies?

A. The standard deviation of number of units sold for company A is less than the standard deviation of number of units sold for company B.

B. The standard deviation of number of units sold for company A is greater than the standard deviation of number of units sold for company B.

C. The standard deviation of number of units sold for company A is equal to the standard deviation of number of units sold for company B.

D. There is not enough information to compare the standard deviations.

---

46 ee670944-e520-4f88-880a-a0158fe35ef8 · Medium

The table shows the frequency of values in a data set.

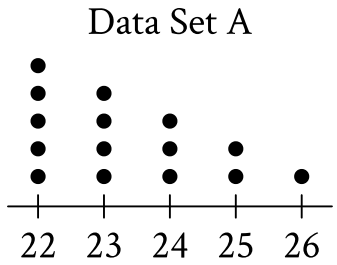
Value	Frequency
19	7
21	1
23	7
25	4

What is the minimum value of the data set?

Student-produced response: _____

---

47 a1b19718-6d73-483b-ac58-8fda7ec4a928 · Medium



- The data for the dot plot are as follows:
  - 22: 5 dots
  - 23: 4 dots
  - 24: 3 dots
  - 25: 2 dots
  - 26: 1 dot

Data set A has 15 values and is represented by the dot plot shown. Data set B is created by adding 46 to each of the values in data set A. Which of the following correctly compares the medians and the ranges of data sets A and B?

A. The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.

B. The median of data set B is greater than the median of data set A, and the range of data set B is greater than the range of data set A.

C. The median of data set B is equal to the median of data set A, and the range of data set B is greater than the range of data set A.

D. The median of data set B is equal to the median of data set A, and the range of data set B is equal to the range of data set A.

---

48 028792-DC · Medium

If a is the mean and b is the median of nine consecutive integers, what is the value of $|a-b|$?

Student-produced response: _____

---

49 070626-DC · Medium

2, 10, 3, 7, 6

The mean of the list of numbers above is what fraction of the sum of the five numbers?

Student-produced response: _____

---

50 026381-DC · Medium

Data set A: 5, 5, 5, 5, 5, 5, 5, 5, 5, 5
Data set B: 5, 5, 5, 5, 5, 5, 5, 5, 5, 100

Which of the following statements about the means and medians of data set A and data set B is true?

A. Only the means are different.

B. Only the medians are different.

C. Both the means and the medians are different.

D. Neither the means nor the medians are different.

---

51 cc8606cf-9fd8-4c61-9cb6-1bcb5b969d36 · Medium

4, 10, 18, 4, 4, 5, 6, 5

What is the median of the data set shown?

A. 4

B. 5

C. 7

D. 14

---

52 2466799c-cc68-42e1-b0cb-461b4ca5597c · Medium

The list gives the mass, in grams, of 5 alpine marmots.

4,010; 4,010; 3,030; 4,050; 3,050

What is the mean mass, in grams, of these 5 alpine marmots?

Student-produced response: _____

---

53 d2d90742-b90d-4d9c-b0ec-3254255f86be · Medium

Data value	Frequency
6	3
7	3
8	8
9	8
10	9
11	11
12	9
13	0
14	6

The frequency table summarizes the 57 data values in a data set. What is the maximum data value in the data set?

Student-produced response: _____

---

54 14851-DC · Medium

For which of the following data sets is the mean greater than the median?

A. 5, 5, 5, 5, 5, 5, 5, 5, 5

B. 0, 10, 20, 30, 40, 50, 60, 70, 80

C. 2, 4, 8, 16, 32, 64, 128, 256, 512

D. 7, 107, 107, 207, 207, 207, 307, 307, 307

---

55 029327-DC · Medium

15, 14, 18, 17, x

The mean and the median of the five numbers above are equal. Which of the following is NOT a possible value of x ?

A. 6

B. 11

C. 16

D. 21

---

56 12877eac-b005-471d-851e-37b10d8a7f90 · Medium

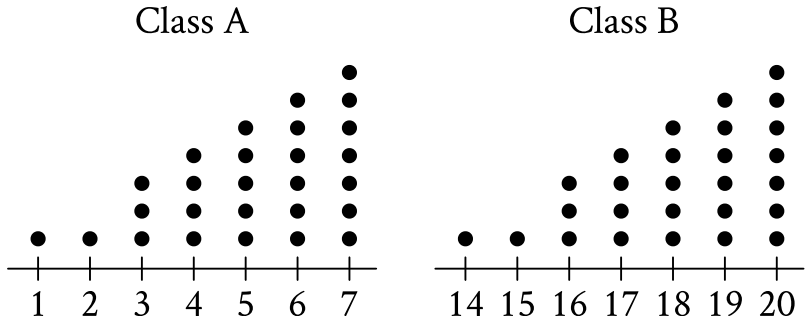
Which of the following lists represents a data set with the smallest standard deviation?

A. 51, 52, 54, 56, 57

B. 52, 52, 54, 56, 56

C. 52, 53, 54, 55, 56

D. 53, 54, 54, 54, 55



- For the dot plot titled Class A:
  - The number line ranges from 1 to 7 in increments of 1.
  - The data for the dot plot are as follows:
    - 1: 1 dot
    - 2: 1 dot
    - 3: 3 dots
    - 4: 4 dots
    - 5: 5 dots
    - 6: 6 dots
    - 7: 7 dots
- For the dot plot titled Class B:
  - The number line ranges from 14 to 20 in increments of 1.
  - The data for the dot plot are as follows:
    - 14: 1 dot
    - 15: 1 dot
    - 16: 3 dots
    - 17: 4 dots
    - 18: 5 dots
    - 19: 6 dots
    - 20: 7 dots

Each of the dot plots shown represents the number of glue sticks brought in by each student for two classes, class A and class B. Which statement best compares the standard deviations of the numbers of glue sticks brought in by each student for these two classes?

A. The standard deviation of the number of glue sticks brought in by each student for class A is less than the standard deviation of the number of glue sticks brought in by each student for class B.

B. The standard deviation of the number of glue sticks brought in by each student for class A is equal to the standard deviation of the number of glue sticks brought in by each student for class B.

C. The standard deviation of the number of glue sticks brought in by each student for class A is greater than the standard deviation of the number of glue sticks brought in by each student for class B.

D. There is not enough information to compare these standard deviations.

---

58 e556a06b-805c-4238-8c14-3bcafe7e1ddf · Medium

$$23, 27, 27, 32, 35, 36, 52$$

What is the range of the 7 scores shown?

Student-produced response: _____

---

59 00566-DC · Hard

The mean score of 8 players in a basketball game was 14.5 points. If the highest individual score is removed, the mean score of the remaining 7 players becomes 12 points. What was the highest score?

A. 20

B. 24

C. 32

D. 36

---

60 b09d507c-3dde-4f24-af93-e07719f875cf · Hard

Data set A consists of the heights of 75 buildings and has a mean of 32 meters. Data set B consists of the heights of 50 buildings and has a mean of 62 meters. Data set C consists of the heights of the 125 buildings from data sets A and B. What is the mean, in meters, of data set C?

Student-produced response: _____

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61 07831-DC · Hard

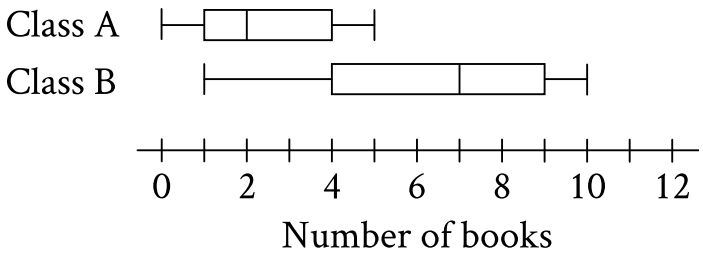
The mean amount of time that the 20 employees of a construction company have worked for the company is 6.7 years. After one of the employees leaves the company, the mean amount of time that the remaining employees have worked for the company is reduced to 6.25 years. How many years did the employee who left the company work for the company?

A. 0.45

B. 2.30

C. 9.00

D. 15.25



- From left to right, the values of the vertical bars in the Class A box plot are as follows:
  - First vertical bar: 0
  - Second vertical bar: 1
  - Third vertical bar: 2
  - Fourth vertical bar: 4
  - Fifth vertical bar: 5
- From left to right, the values of the vertical bars in the Class B box plot are as follows:
  - First vertical bar: 1
  - Second vertical bar: 4
  - Third vertical bar: 7
  - Fourth vertical bar: 9
  - Fifth vertical bar: 10

The two box plots show the distribution of number of books read over the summer by the students in two different English classes. What is the positive difference between the ranges of number of books read over the summer for the two classes?

Student-produced response: _____

63 07857-DC · Hard

Percent of Residents Who Earned a Bachelor's Degree or Higher

State	Percent of residents
State A	21.9%
State B	27.9%
State C	25.9%
State D	19.5%
State E	30.1%
State F	36.4%
State G	35.5%

A survey was given to residents of all 50 states asking if they had earned a bachelor's degree or higher. The results from 7 of the states are given in the table above. The median percent of residents who earned a bachelor's degree or higher for all 50 states was 26.95%. What is the difference between the median percent of residents who earned a bachelor's degree or higher for these 7 states and the median for all 50 states?

A. 0.05%

B. 0.95%

C. 1.22%

D. 7.45%

---

64 070627-DC · Hard

	Masses (kilograms)					
Andrew	2.4	2.5	3.6	3.1	2.5	2.7
Maria	x	3.1	2.7	2.9	3.3	2.8

Andrew and Maria each collected six rocks, and the masses of the rocks are shown in the table above. The mean of the masses of the rocks Maria collected is 0.1 kilogram greater than the mean of the masses of the rocks Andrew collected. What is the value of x ?

Student-produced response: _____

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65 1f3a2144-9cce-41ef-bcb2-8dfc5cc97d8b · Hard

Data set A consists of 10 positive integers less than 60. The list shown gives 9 of the integers from data set A.

$$43, 45, 44, 43, 38, 39, 40, 46, 40$$

The mean of these 9 integers is 42. If the mean of data set A is an integer that is greater than 42, what is the value of the largest integer from data set A?

Student-produced response: _____

Weight (pounds)	Frequency
13	12
14	8
15	5
16	7
17	9
18	10
19	13
20	7

The frequency table summarizes a data set of the weights, rounded to the nearest pound, of 71 tortoises. A weight of 39 pounds is added to the original data set, creating a new data set of the weights, rounded to the nearest pound, of 72 tortoises. Which statement best compares the mean and median of the new data set to the mean and median of the original data set?

A. The mean of the new data set is greater than the mean of the original data set, and the median of the new data set is greater than the median of the original data set.

B. The mean of the new data set is greater than the mean of the original data set, and the medians of the two data sets are equal.

C. The mean of the new data set is less than the mean of the original data set, and the median of the new data set is less than the median of the original data set.

D. The mean of the new data set is less than the mean of the original data set, and the medians of the two data sets are equal.

67 4135b77c-4178-4cdc-8f4a-8f46e634b934 · Hard

Data set F consists of 55 integers between 170 and 290. Data set G consists of all the integers in data set F as well as the integer 10. Which of the following must be less for data set F than for data set G?

I. The mean
II. The median

A. I only

B. II only

C. I and II

D. Neither I nor II

---

68 028957-DC · Hard

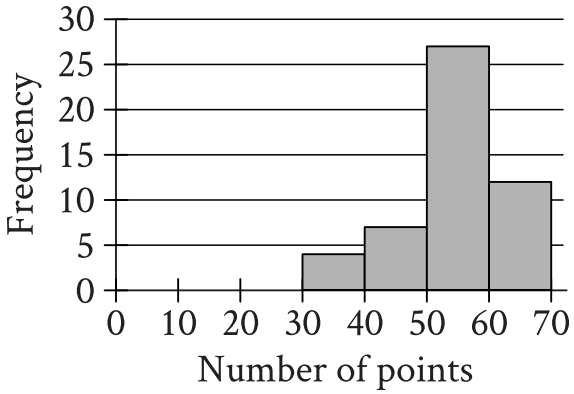
A data set of 27 different numbers has a mean of 33 and a median of 33. A new data set is created by adding 7 to each number in the original data set that is greater than the median and subtracting 7 from each number in the original data set that is less than the median. Which of the following measures does NOT have the same value in both the original and new data sets?

A. Median

B. Mean

C. Sum of the numbers

D. Standard deviation



- The histogram has 4 bins.
- The frequency values of the 4 bins are as follows:
  - 30 to 39 points: More than halfway between 0 and 5
  - 40 to 49 points: About halfway between 5 and 10
  - 50 to 59 points: About halfway between 25 and 30
  - 60 to 69 points: About halfway between 10 and 15

The histogram summarizes data set A, which represents the number of points per player earned by 50 players of a game. A new player earns 18 points playing the game, and this number of points is added to data set A to create data set B with 51 values. Which of the following must be true?

I. The median number of points per player for data set B is less than the median number of points per player for data set A.
II. The mean number of points per player for data set B is less than the mean number of points per player for data set A.

A. I only

B. II only

C. I and II

D. Neither I nor II

70 029610-DC · Hard

Value	Frequency
1	a
2	2a
3	3a
4	2a
5	a

The frequency distribution above summarizes a set of data, where *a* is a positive integer. How much greater is the mean of the set of data than the median?

A. 0

B. 1

C. 2

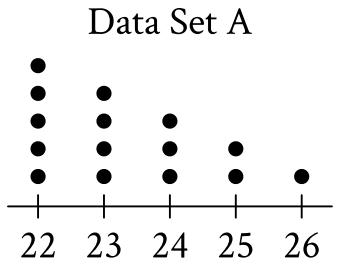
D. 3

---

71 b9761b18-7942-4214-89a5-1726fe26fe75 · Hard

Data set A consists of the heights of 75 objects and has a mean of 25 meters. Data set B consists of the heights of 50 objects and has a mean of 65 meters. Data set C consists of the heights of the 125 objects from data sets A and B. What is the mean, in meters, of data set C?

Student-produced response: _____



- The data for the dot plot are as follows:
  - 22: 5 dots
  - 23: 4 dots
  - 24: 3 dots
  - 25: 2 dots
  - 26: 1 dot

The dot plot represents the 15 values in data set A. Data set B is created by adding 56 to each of the values in data set A. Which of the following correctly compares the medians and the ranges of data sets A and B?

A. The median of data set B is equal to the median of data set A, and the range of data set B is equal to the range of data set A.

B. The median of data set B is equal to the median of data set A, and the range of data set B is greater than the range of data set A.

C. The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.

D. The median of data set B is greater than the median of data set A, and the range of data set B is greater than the range of data set A.

73 070912-DC · Hard

For a certain computer game, individuals receive an integer score that ranges from 2 through 10. The table below shows the frequency distribution of the scores of the 9 players in group A and the 11 players in group B.

Score	Score Frequencies	
	Group A	Group B
2	1	0
3	1	0
4	2	0
5	1	4
6	3	2
7	0	0
8	0	2
9	1	1
10	0	2
Total	9	11

The median of the scores for group B is how much greater than the median of the scores for group A?

Student-produced response: _____

74 32297287-d739-4961-acc3-2eed761e3e0d · Hard

Each of the following frequency tables represents a data set. Which data set has the greatest mean?

A.

Value	Frequency
70	4
80	5
90	6
100	7

B.

Value	Frequency
70	6
80	6
90	6
100	6

C.

Value	Frequency
70	7
80	6
90	6
100	7

D.

Value	Frequency

70	8
80	5
90	5
100	8

---

75 a97f29fd-deb9-4f76-baff-4f4bea70e929 · Hard

A scientist measured the lengths of 240 gray seals from Muskeget Island and 120 gray seals from Sable Island. The scientist determined that the mean length of the 240 gray seals from Muskeget Island was 88 inches and the mean length of the 120 gray seals from Sable Island was 94 inches. What was the mean length of all 360 gray seals the scientist measured for this study?

A. 89

B. 90

C. 91

D. 92

---

76 45c46b40-e7e6-4f3e-bbb1-22e0862c3911 · Hard

Value	Data set A frequency	Data set B frequency
30	2	9
34	4	7
38	5	5
42	7	4
46	9	2

Data set A and data set B each consist of 27 values. The table shows the frequencies of the values for each data set. Which of the following statements best compares the means of the two data sets?

A. The mean of data set A is greater than the mean of data set B.

B. The mean of data set A is less than the mean of data set B.

C. The mean of data set A is equal to the mean of data set B.

D. There is not enough information to compare the means of the data sets.

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77 f98d0c41-892c-4afb-82e0-d348eba42052 · Hard

$$a, 26, 29, b, 31, 47, c$$

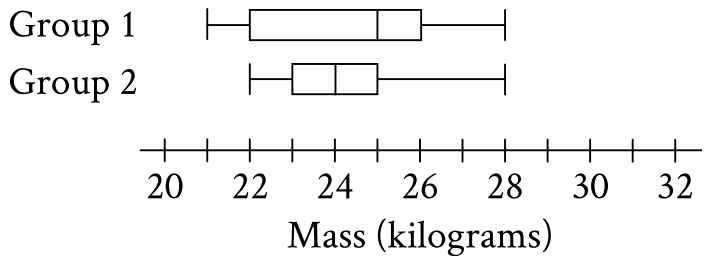
For the given data set, the data values are listed in ascending order, where a, b, and c are constants. For this data set, the mean is 36, the median is 29, and the range is 72. What is the value of c?

A. 54

B. 72

C. 81

D. 98



- From left to right the values of the vertical bars in the Group 1 box plot are as follows:
  - First vertical bar: 21
  - Second vertical bar: 22
  - Third vertical bar: 25
  - Fourth vertical bar: 26
  - Fifth vertical bar: 28
- From left to right the values of the vertical bars in the Group 2 box plot are as follows:
  - First vertical bar: 22
  - Second vertical bar: 23
  - Third vertical bar: 24
  - Fourth vertical bar: 25
  - Fifth vertical bar: 28

The box plots summarize the masses, in kilograms, of two groups of gazelles. Based on the box plots, which of the following statements must be true?

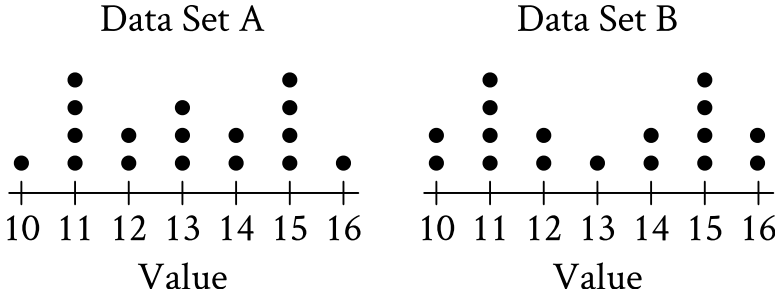
A. The mean mass of group 1 is greater than the mean mass of group 2.

B. The mean mass of group 1 is less than the mean mass of group 2.

C. The median mass of group 1 is greater than the median mass of group 2.

D. The median mass of group 1 is less than the median mass of group 2.

The dot plots represent the distributions of values in data sets A and B.



- For the dot plot titled Data Set A:
  - The number line ranges from 10 to 16 in increments of 1.
  - The data for the dot plot are as follows:
    - 10: 1 dot
    - 11: 4 dots
    - 12: 2 dots
    - 13: 3 dots
    - 14: 2 dots
    - 15: 4 dots
    - 16: 1 dot
- For the dot plot titled Data Set B:
  - The number line ranges from 10 to 16 in increments of 1.
  - The data for the dot plot are as follows:
    - 10: 2 dots
    - 11: 4 dots
    - 12: 2 dots
    - 13: 1 dot
    - 14: 2 dots
    - 15: 4 dots
    - 16: 2 dots

Which of the following statements must be true?

I. The median of data set A is equal to the median of data set B.
II. The standard deviation of data set A is equal to the standard deviation of data set B.

A. I only

B. II only

C. I and II

D. Neither I nor II

---

80 46b54ee4-c738-4f84-acf0-ed71e853d400 · Hard

Data Set Q

Value	Frequency
a	110
b	30
c	0

Data Set R

Value	Frequency
a	0
b	3
c	11

Data sets Q and R are summarized in the tables shown, where a, b, and c are positive, consecutive integers and $a < b < c$. Which statement comparing the means of the data sets is true?

A. The mean of data set R is $\frac{211}{14}$ less than the mean of data set Q.

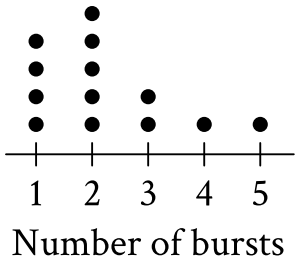
B. The mean of data set R is $\frac{211}{14}$ greater than the mean of data set Q.

C. The mean of data set Q is 10 times the mean of data set R.

D. The means of the two data sets are equal.

---

81 84fea272-2bed-48c4-9ffd-dfe5cc5edb2f · Hard



- The data for the dot plot are as follows:
  - 1: 4 dots
  - 2: 5 dots
  - 3: 2 dots
  - 4: 1 dot
  - 5: 1 dot

The dot plot represents a data set of the number of bursts for 13 eruptions of a steam vent. If an additional eruption with 11 bursts is added to this data set to create a new data set of 14 eruptions, which of the following measures will be greater for the new data set than for the original data set?

I. The median number of bursts
II. The mean number of bursts

A. I and II

B. I only

C. II only

D. Neither I nor II

82 27d95f70-58ff-4b21-92f7-c74df5011b3a · Hard

The table shown summarizes the number of employees at each of the 17 restaurants in a town.

Number of employees	Number of restaurants
2 to 7	2
8 to 13	4
14 to 19	2
20 to 25	7
26 to 31	2

Which of the following could be the median number of employees for the restaurants in this town?

A. 2

B. 9

C. 15

D. 21

83 33c49cb1-9a61-48d8-8b47-83909effa87d · Hard

For data set A, the table summarizes the distribution of the number of deliveries received by an office each day during a period of 11 days.

Deliveries	Days
0	2
3	2
4	2
5	2
6	1
7	1
13	1

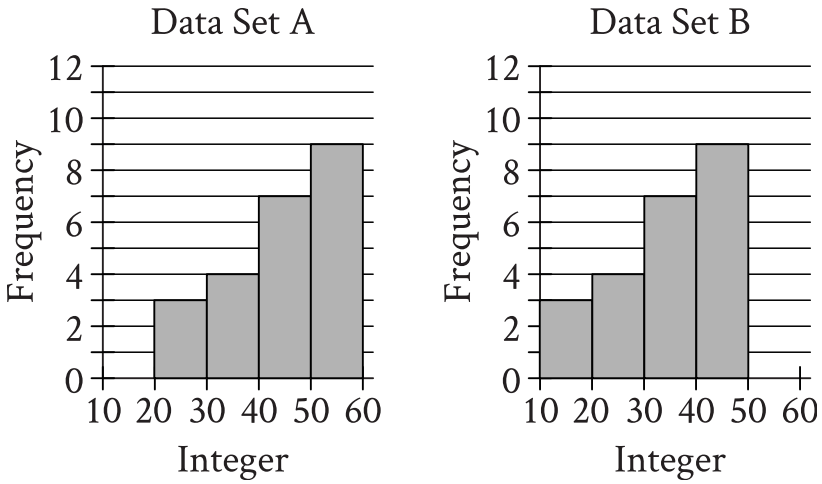
The data value 13 was recorded in error and is removed from data set A to create data set B, which consists of the remaining 10 data values. Which statement best compares the median of data set A and the median of data set B?

A. The median of data set B is less than the median of data set A.

B. The median of data set B is greater than the median of data set A.

C. The median of data set B is equal to the median of data set A.

D. There is not enough information to compare the medians of the two data sets.



- Data Set A:
  - The horizontal axis is labeled Integer. It ranges from 10 to 60 and is divided into 5 equal intervals.
  - The vertical axis is labeled Frequency. It ranges from 0 to 12 in increments of 1, with values marked every 2 grid lines.
  - The histogram has a skewed right shape.
  - The histogram has 4 bins.
  - The Frequency data for the 4 bins are as follows:
    - 20 to 30: 3
    - 30 to 40: 4
    - 40 to 50: 7
    - 50 to 60: 9
- Data Set B:
  - The horizontal axis is labeled Integer. It ranges from 10 to 60 and is divided into 5 equal intervals.
  - The vertical axis is labeled Frequency. It ranges from 0 to 12 in increments of 1, with values marked every 2 grid lines.
  - The histogram has a skewed right shape.
  - The histogram has 4 bins.
  - The Frequency data for the 4 bins are as follows:
    - 10 to 20: 3
    - 20 to 30: 4
    - 30 to 40: 7
    - 40 to 50: 9

Two data sets of 23 integers each are summarized in the histograms shown. For each of the histograms, the first interval represents the frequency of integers greater than or equal to 10, but less than 20. The second interval represents the frequency of integers greater than or equal to 20, but less than 30, and so on. What is the smallest possible difference between the mean of data set A and the mean of data set B?

A. 0

B. 1

C. 10

D. 23